

INSPECTION TEST PLAN (ITP)								
MANUFACTURER:- THERMOPAC PROCESS ENGG. LLP (CIN AAD-9858) L4,405,THE SUMMIT BUSINESS BAY, WESTERN EXPRESS ,HIGHWAY,VILEPARLE(EAST) , MUMBAI 400057, MAHARASHTRA, INDIA		CUSTOMER:- Waste Petroleum Combustion Limited PTY. LTD. 39 Subway Road Pukekohe, AUCKLAND, 2120 New Zealand		PROJECT:	WPC Recycling Plant UOR-4500	EQUIPMENT	WATER NAPHTHA SEPARATOR (RF/SP/E1)	
				PO NO.	66612	DRAWING NO	4823002004001002	
				ITP NO	15	REVISION	02	
				ITP DATE	18-07-2024	Design Verification No & Date	21565257 Issue 1 DT.31/05/2024	
				ITP REVISION	00	QTY	1	
				NOTIFIED BODY	SGS INDIA	Hazard Level	B	
SL.NO	INSPECTION ACTIVITY / ITEM	CHARACTERISTICS CHECKED	REFERENCE DOCUMENTS	ACCEPTANCE CRITERIA	FORMAT OF RECORDS	INSPECTION BY		REMARKS
1	2	3	4	5	6	7		8
1	REVIEW OF DOCUMENTS					M	SGS	
1.1	Design & Drawings	Review & approval	Design & drawing.	Design Verification by notified body	Design Verification certificate	P	R / H	DV certificate No.21565257
1.2	Approval of Inspection Test Plan (ITP)	Review & approval	Drawing, EN13445-3&5, ACOP	compliance to Drawing, EN13445-3 & 5, ACOP	Inspection Test Plan	P	R / A	
1.3	WPS/PQR/WPQ & weld plan	Review & approval	WPS,PQR,WPQ & Weld plan	Compliance to EN287(EN 9606) & EN288, Weld Sketch	WPS/PQR/WPQ & weld plan	P	W / A	EN 287 REPLACED BY EN 9606 for existing document Scope of TPIA us review and for new qualification to be done under SGS Witness.
1.4	Particular Material Appraisal	Review & approval	EN13445-3	Compliance to reference document	AS PER MANUFACTURERS FORMAT	P	R	Review of a notified body
1.5	Pneumatic & Hydrostatic leak test procedures.	Review & approval	Drawing, ASME SECT. VIII DIV.I & test procedure	Compliance to reference document	Approved procedure.	P	R	
1.6	Non destructive examination (NDE) procedure	Review & approval	ASME SECT.V & EN 13445-5	Compliance to reference document	Evaluation report from level III qualified personal as per ISO9712	P	R	
2	MATERIALS							
2.1	Material for pressure parts such as Shell , Dished end , RF pad etc.	Physical , Chem.,dimensional & visual.	Drawing , ASME SECT.II,EN10204	Compliance to specification of drawing & ASME standards.	IR / MTC (3.1)/ Lab Test(3.2)	P	R / W	Check testing to be carried out under SGS witness in NABL accredited Lab in case of original 3.1 MTC is unavailable and issue 3.2 as per EN10204
2.2	Pipes for all Nozzles.	Physical , Chem.,dimensional & visual	Drawing , ASME SECT.II,EN10204	Compliance to specification of drawing & ASME standards.	IR / MTC (3.1)/ Lab Test(3.2)	P	R / W	Check testing to be carried out under SGS witness in NABL accredited Lab in case of original 3.1 MTC is unavailable and issue 3.2 as per EN10204
2.3	All flanges.	Physical , Chem.,dimensional & visual	Drawing , ASME SECT.II, EN 13445-3	SA105 ,EN13445-3	IR / MTC	P	R / V	Original 3.1 certificate required
2.4	Lugs. supports etc.	Physical , Chem.,dimensional & visual	Drawing , ASME SECT.II,EN10204	Compliance to specification of drawing & ASME standards.	IR / MTC (3.1)/ Lab Test(3.2)	P	R / W	Check testing to be carried out under SGS witness in NABL accredited Lab in case of original 3.1 MTC is unavailable and issue 3.2 as per EN10204
2.5	Hard Ware (bolts,nuts etc.) & Gaskets	Dimensional./ Visual	Drawing,ASME SECT.II,EN13445-3	Compliance to SA-193 B7,B8 /SA-194 B8 /SA-194 2H/IS2062 & AF159	IR / MTC	P	R	
2.6	Welding consumables	Review	ASME SECT.II C & respective WPS	Compliance to requirement of WPS/Welding process	IR / MTC	P	R	Manufacturer's MTC

3	IN PROCESS INSPECTION							
3.11	Identification of material and witnessing stamp transfer of identification marks in manufacturer's works	Identification of heat no ,w.r.t Marking TC	Approved Drawing,cutting layout,IR	Approved Drawing,cutting layout,IR	Joint Inspection report	P	W	
3.12	Examination of material cut edges and heat affected Zones	Visual	Approved Drawing , ACOP & 13445-4	Compliance to Drawing , ACOP & 13445-4	IR	P	R	
3.13	Examination of formed part such as dish, cone etc	Dimensional./ Visual	Approved Drawing / EN13445-4,5	Compliance to Approved Drawing / EN13445-4,5	IR	P	H	
3.14	Welded blanks for shell & dish, cone etc	Dimensional./ Visual	Approved Drawing / EN13445-4,5	Compliance to Approved Drawing / EN13445-4,5	IR	P	H	
3.15	Examination of set up of seams for welding including dimensional check, examination of weld preparation, tack welds shell ,dish, cone, other pressure pressure parts etc	Dimensional./ Visual	Approved Drawing / EN13445-4,5	Compliance to Approved Drawing / EN13445-4,5/ ACOP	IR	P	H	
3.16	Inspection of second side of weld preparations after first side is completed and root cleaned	Visual	Respective WPS	Compliance to Approved Drawing	IR	P	R / W	
3.17	Nozzle, manhole Set-up on Shell & Dish End Other Pressure parts attachment to Pressure Vessel in tack weld condition	Visual, Dimensional,	Approved drawing,	Compliance to Approved Drawing /	IR	P	H	
3.18	NDT tests for weld joints.	Weld defects	Approved drawing,weld plan,NDT procedures	No weld defects	NDT Reports & Films	P	R / H	ISO9712 or equivalent NDT level II / III personal to be review RT films & Reports
3.19	Final Visual & Dimension Inspection	Dimensional./ Visual	Approved drawing	Compliance to Approved Drawing dimensions	IR	P	W	
3.20	Pneumatic Test of Nozzle RF pads	Weld leak	Approved drawing &	No leaks.	IR	P	R / W	
3.22	Hydrostatic Test	Leak test	Approved drawing &	No leaks.	IR	P	W	
4	FINAL ASSESSMENT							
4.1	Documentation review (heat chart,weld plans,test reports, NDT reports etc.)	Documentation requirement as per PO	Customer PO, EN13445-5,ACOP	Compliance to Customer PO, EN13445-5,ACOP	Declaration of compliance	P	R	
4.2	Buffing.	Mirror Finishing	As per Procedure	Compliance to procedure	IR	P	R	
4.3	Stamping of data plate & signing of declaration.	Final acceptance of vessel	Assessment of documents as per EN13445-5 & ACOP	Compliance to Final assessment of the vessel.	Declaration of compliance with the standard.		P	
			LEGEND:					
			M:- MANUFACTURER/SUB-SUPPLIER					
			C:- CUSTOMER					
			SGS:- APPROVED INSPECTION AGENCY					
			RW - Random witness					
			P:- PERFORM					
			W :- WITNESS					
H :- HOLD								
FORMAT NO.:- ITP_WPC_Pressure Parts		PREPARED BY:		V :- VERIFICATION		APPROVED BY		